

Engine Does Not Shut Down with High Coolant Temperature

Symptoms

- The SDU410 will **not** shut down the engine if high coolant temperature exists.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SDU410 input signals are switches. These switches are normally open and closed to activate a shutdown.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check customer interface box	
	STEP 1A. Check the customer interface box logic unit LED illumination.	Are any alarms active or LEDs illuminated?
	STEP 1B. Check the SDU410 power supply wire for +24-VDC.	Less than +24-VDC?
STEP 2.	Check customer interface box wiring	
	STEP 2A. Check coolant temperature signal and return wires for an open.	Less than 10 ohms?
STEP 3.	Check the OEM wiring harness	
	STEP 3A. Check coolant temperature signal and return wires for an open.	Less than 10 ohms?

STEP 1. Check customer interface box

STEP 1A. Check the customer interface box logic unit LED illumination.

Conditions :

- Check the DCU410 unit for alarms and LED illumination.

Action	Specification/Repair	Next Step
Check for alarms and LED illumination on the DCU410 unit.	Are any alarms active or LEDs illuminated? YES	Contact a Cummins® Authorized Repair Location
	Are any alarms active or LEDs illuminated? NO	1B

STEP 1B. Check the DCU410 power supply wire for +24-VDC.

Conditions :

- Open the customer interface box

Action	Specification/Repair	Next Step
Check the voltage at the shutdown unit supply 24-VDC at the SDU410 unit. • Place one test lead on the shutdown unit supply 24-VDC supply wire at the SDU410 unit. • Place the other test lead on the shutdown unit return wire at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for connection pin identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete
	Less than +24-VDC? NO	2A

STEP 2. Check customer interface box wiring

STEP 2A. Check the coolant temperature signal and return wires for an open.

Conditions :

- Open the customer interface box
- Disconnect customer interface box to OEM harness cable connector C4 from the customer interface box.
- Disconnect OEM harness 23-pin C2 connector from the customer interface box.

Action	Specification/Repair	Next Step
Check the coolant temperature signal and return wires for an open. • Place one test lead on the coolant temperature signal wire at the SDU410 unit. • Place the other test lead on the coolant temperature signal pin at the C4 connector. • Place one test lead on the coolant temperature return wire at the SDU410 unit. • Place the other test lead on the coolant temperature return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for connection pin identification.	Less than 10 ohms? YES	3A
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the SDU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 3. Check the OEM wiring harness

STEP 3A. Check the coolant temperature signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect customer interface box to OEM harness cable connector C4 from the customer interface box.
- Disconnect the OEM harness connector at the OEM connector on engine.

Action	Specification/Repair	Next Step
Check the coolant temperature signal wire for an open. • Place one test lead on the coolant temperature signal wire at the C4 connector. • Place the other test lead on the coolant temperature signal wire at the C11 connector. • Place one test lead on the coolant temperature return wire at the C4 connector. • Place the other test lead on the coolant temperature return wire at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for connection pin identification.	Less than 10 ohms? YES	Repair complete
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the OEM wiring harness. Refer to the OEM service manual.	Contact a Cummins® Authorized Repair Location